# 1. Identity

| | |
|---|---|
| Document | STAR-PKG-STAR-MANEUVER-EN · v1.4 · 2026-10-20 |
| Package | star-maneuver |
| Version | 0.2.2 |
| Capability | star.orbit.maneuver |
| Maturity | VERIFIED |
| Python | ⩾3.9 |
| Source commit | 94fcf7c |
| Source | https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_maneuver |

# 2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

# 3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

# 4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_maneuver-0.2.2-py3-none-any.whl
```

| FILE | SHA-256 |
|---|---|
| star_maneuver-0.2.2-py3-none-any.whl | 2105bf37e63d26ee39181929cc4e07deeca4eebbe5bbc8bb0e8043d01ebdb45c |
| star_maneuver.py | c1908a47917dfdba2b61883b2d442ec547e51c6bc4986e94bb71d4ef7d8dfc83 |
| star_maneuver-0.2.2-source.zip | f4557fc2857bb1b06d97fcfa27bdb0fc89c860683efe980d967cb3a83a689650 |

# 5. What it does, and its limits

Impulsive manoeuvre arithmetic between circular coplanar orbits and plane changes. Standard library only.

Finite burns, gravity losses, non-coplanar Hohmann with optimal plane-change split, any flight qualification.

- `vis_viva(r, a, mu)`; `hohmann(r1, r2, mu)` and `bielliptic(r1, r2, rb, mu)` return per-burn and total delta-v (km/s) and time of flight (s). Units: km, s, km/s, km^3/s^2 (default mu = 398600.4418, IERS Conventions 2010).
- `plane_change(v, di)` = 2 v sin(di/2); `combined_dv(v1, v2, di)` by the law of cosines (radians).
- Invalid input fails explicitly with `ValueError` (non-positive radii, rb < max(r1, r2), negative speed).

# 6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

# 7. Citation and bill of materials

```
@software{cavazzini_star_maneuver_0_2_2,
  author  = {Cavazzini, Andrea},
  title   = {{star-maneuver}: Closed-form vis-viva and Hohmann orbital transfer engine},
```

```
  version = {0.2.2},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_maneuver/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

# 8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts